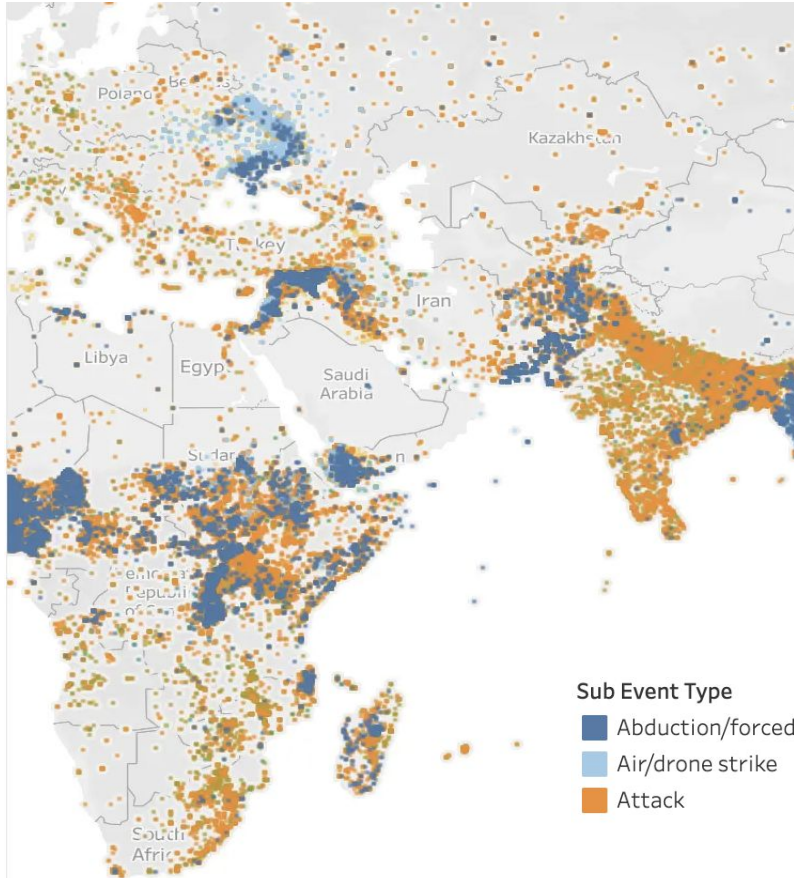


**Data Visualization Pitch**  
**Student Individual Assignment**



# The Human Cost of Civilian Targeting

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## Research questions

How has violence against civilians changed globally between 2021 and 2025, and where has its human cost been most severe?



Geographic  
Concentration

Which regions record the highest number of civilian-targeting events?



Unequal human  
cost

How does the human cost of civilian-targeting vary across regions?



Drivers of  
Human Loss

Which forms drive fatalities in the most affected regions??



Areas of Greatest  
Severity

Which countries show the greatest need for attention from a civilian-protection perspective?

# About the Data

ACLED - Armed Conflict Location & Event Data: <https://acleddata.com/>

ACLED is a global event-based database that records political violence and protest events. Each event includes information on **date, location, actors involved, event type, civilian targeting and reported fatalities**. Data are collected mainly from media, local sources and partner organizations.

## Selected Dataset:

For this project, the **global ACLED** dataset was filtered to cover the period **2021-2025** and to include events where the **civilian\_targeting** variable identifies civilians as the **main or sole target of violence**. It also captures cases across different event types, such as **Explosions/Remote violence, Riots** and **Excessive force against protesters**.

## Data Quality and Limitations:

ACLED applies **standardized coding rules** and **quality checks**, but its data still depend on reported sources, so some events may be **underreported** or **documented unevenly** across countries and conflicts. Fatality figures are also estimates and should be interpreted as **reported deaths** rather than exact counts. A further limitation is that **2025 is a partial year**, with data available only until August 25. This limitation is explicitly considered when interpreting temporal trends.

## Licence and Access:

ACLED data are not fully open for unrestricted redistribution. To obtain the complete dataset required for this analysis, it was necessary to **register** with ACLED and **request access** to its data platform. The data can be used for this **non-commercial academic project** under ACLED's **End User License Agreement (EULA)**.

# Methodology

**Data Collection:** The dataset was **downloaded** as a **CSV** from the official ACLED Data Export platform.

## Data Processing & Cleaning:

Processing was carried out using **Python** with:

- **Pandas** for data cleaning and filtering
- **NumPy** for numerical calculations
- **Matplotlib** for base visualization and charts
- **Scipy.stats** for statistical analysis
- **Seaborn** for statistical visualization

The cleaning process included:

- checking missing values and duplicate records
- converting numeric and date variables to the correct format
- keeping only variables needed for the analysis
- checking consistency of geographic and event-related fields

## Data Transformation:

The selected dataset was aggregated by **time, geography and event type** according to the research questions. Key indicators were then calculated, including the **number of events, total fatalities and average fatalities per event**, used as a measure of lethality. For the country comparison, **event frequency** and **lethality** were **normalized** to a common scale and combined with equal weight to create the **Relative Attention Index**, highlighting countries with high values on both dimensions.

# Methodology

## Analytical and Visual Techniques:

- **Tools:** Python and Tableau
- **Descriptive analysis:** to summarize events and fatalities
- **Temporal analysis:** to compare changes from 2021 to 2025, treating 2025 as a partial year
- **Comparative analysis:** to compare regions, countries, event types, and actors
- **Lethality analysis:** to measure severity using average fatalities per event
- **Spatial analysis:** to explore the geographic distribution of events
- **Composite indicator analysis:** to combine event frequency and lethality for the final country comparison

## Tools and Reproducibility:

- **AI Integration:** Chatgpt was used for code optimization of complex data visualizations and debugging.
- **Reproducibility:** all data processing and analysis steps are documented in a Jupyter Notebook.
- **Source Code:** The code used for cleaning, analysis and visualizations is available on **GitHub** (View Documentation [HERE](#))
- **Data Source:** project can be reproduced by users with access to the original **ACLED dataset**.

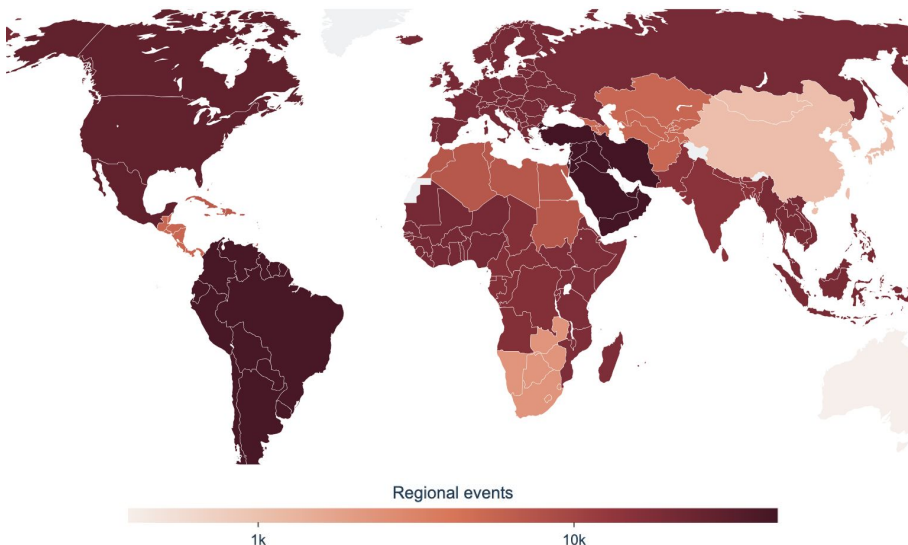
# Insights from the Data

## GLOBAL OVERVIEW

### Insight 1: Civilian Harm is Concentrated

Civilian-targeting events are unevenly distributed across regions and countries.

Civilian-targeting events, 2021–2025  
2021–2024 full years | 2025 partial through 25 August



## MAIN FINDINGS

### Insight 2: More events do not mean more deaths

High event counts do not always correspond to the highest fatality tolls.

### Insight 3: Some forms of violence account for more deaths

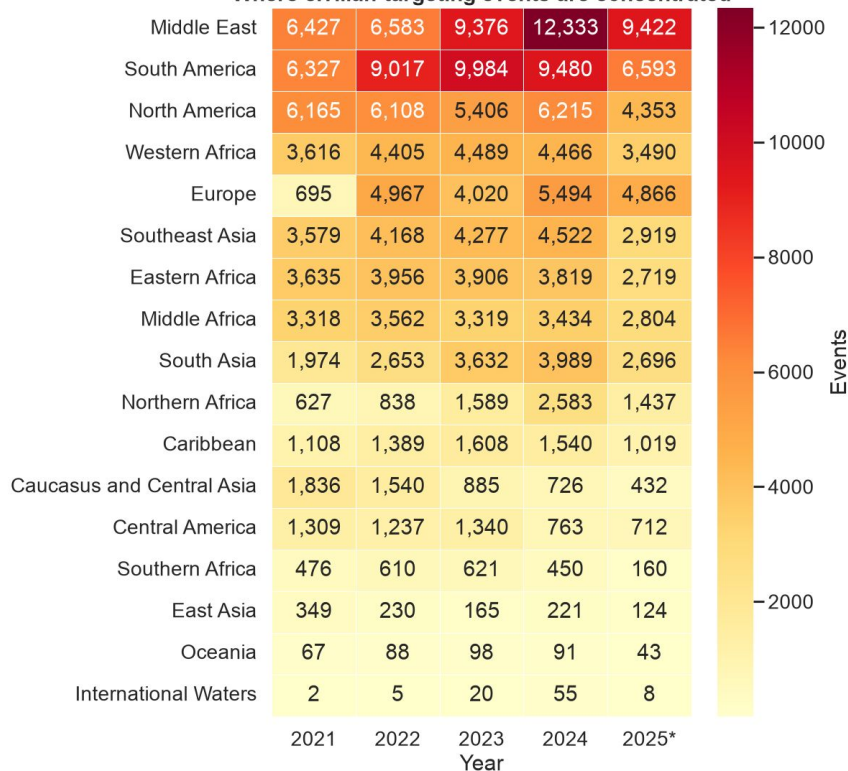
Different event types are associated with very different numbers of reported fatalities.

### Insight 4: Some countries face both high frequency and high lethality

The countries with the highest relative attention are those where frequent civilian-targeting events are combined with high human cost.

# Civilian-targeting is most concentrated in the Middle East

Where civilian-targeting events are concentrated



Civilian-targeting is unevenly distributed across regions, with most events concentrated in a small number of areas.

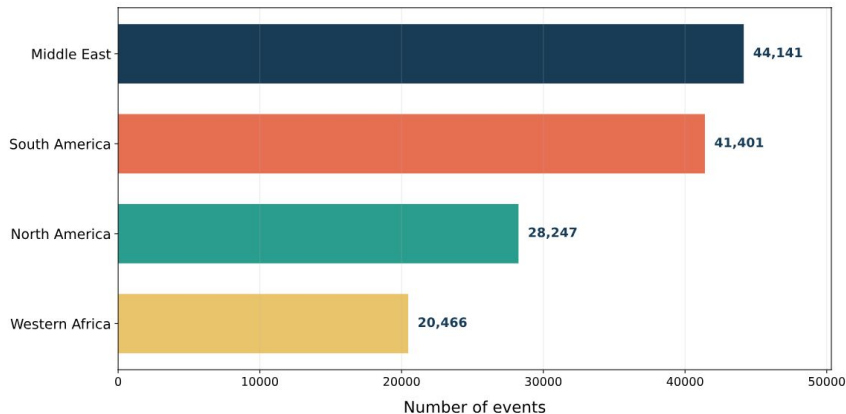
The **Middle East** records the **highest overall number of events**, with about **44,100**, followed closely by **South America** with around **41,400**.

Both regions remain among **those recording the highest event counts across the period shown (2021-2025)**, despite year-to-year variation.

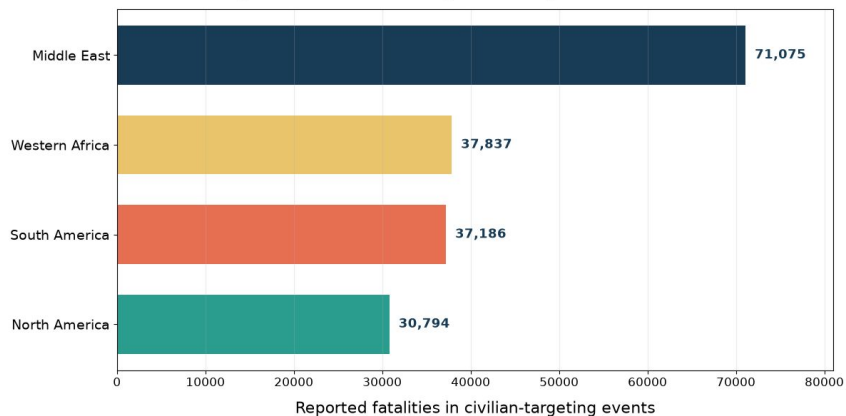
However, the regions with the most events are not always those with the highest human losses.

# How does the human cost of civilian-targeting vary across regions?

Regions with the most civilian-targeting events



Regions with the highest reported fatalities

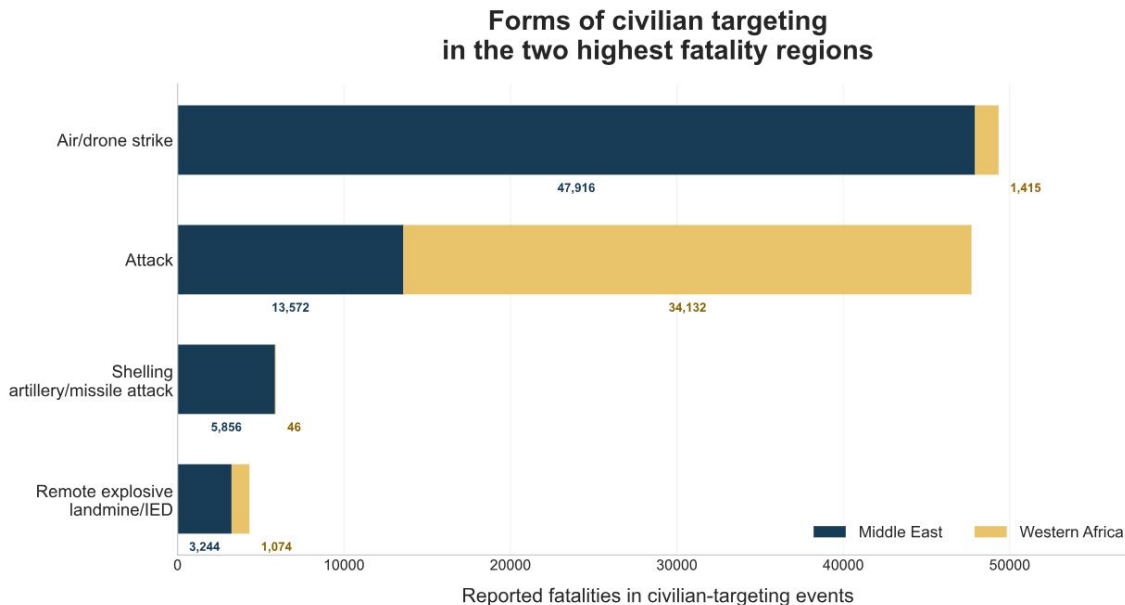


Frequency explains only part of the human burden.

The regional ranking changes when the focus shifts from events to fatalities. The **Middle East remains the most affected region**, with more than 71,000 reported fatalities, but **Western Africa rises sharply**, with more than 37,000 reported fatalities, **despite having far fewer events** than South America or North America.

Human cost depends not only on how often violence occurs, but also on the form it takes. This makes it important to examine which types of civilian-targeting are associated with the highest reported fatalities.

## Which forms drive fatalities in the most affected regions?



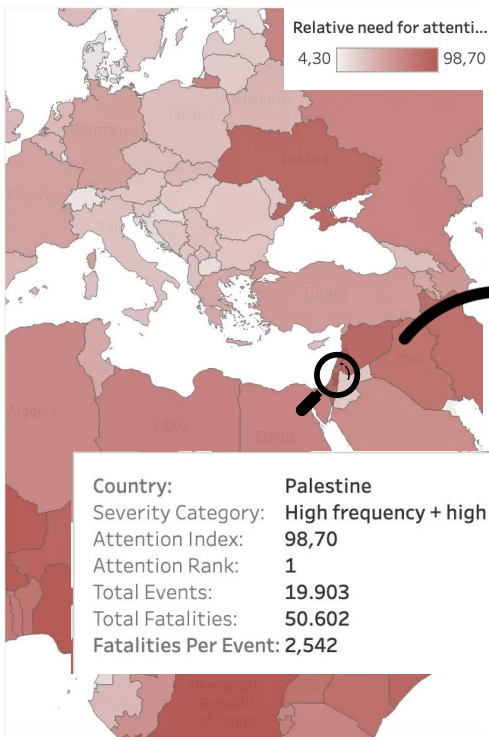
The forms of civilian-targeting associated with the highest reported fatalities vary across regions. In the **Middle East**, **air/drone strikes dominate**, accounting for by far the largest number of reported fatalities (47,916).

In **Western Africa**, **attacks** are instead the main source of **human loss** (34,132), while air/drone strikes and shelling play a much smaller role. This shows that the composition of civilian-targeting violence is not the same across the two regions.

These differences show that the human cost of civilian-targeting is driven by different forms of violence across regions. The final step is therefore to identify where **repeated civilian-targeting and severe human losses overlap at the country level**.

# Which countries show the greatest need for attention from a civilian protection perspective?

GLOBAL VIEW



EVENT MAP



An **interactive dashboard** is available to explore country-level patterns and compare the **Relative Attention Index** across the countries included in the analysis. Higher values indicate a stronger combination of the two indicators used in the index.

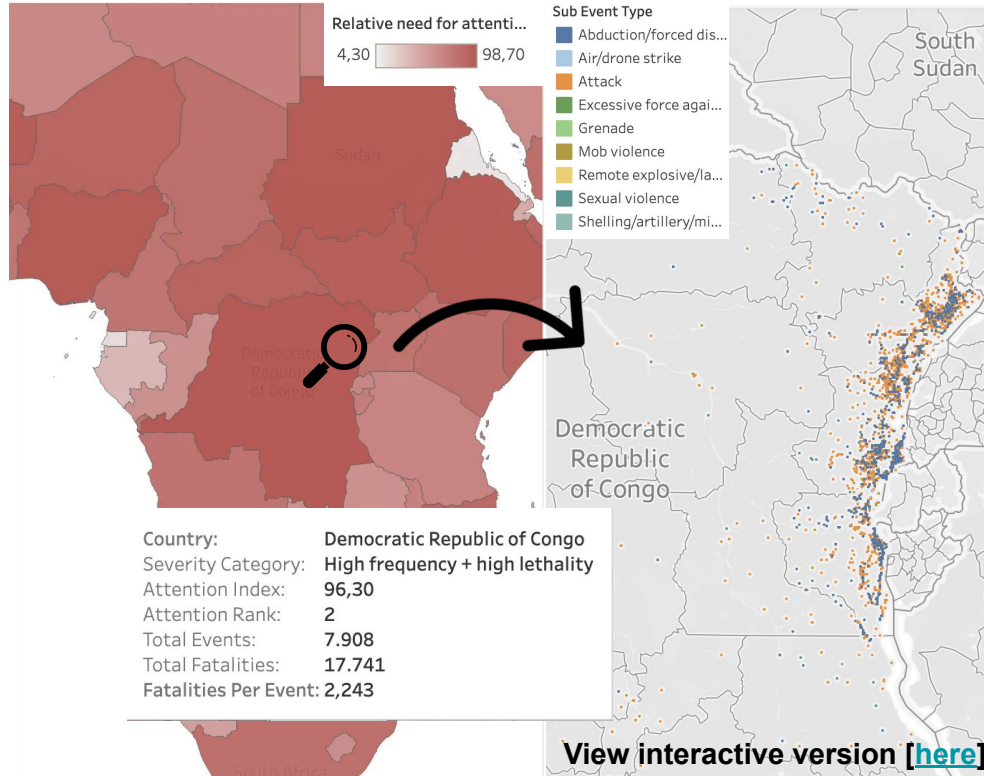
Among the countries analyzed, **Palestine** stands out for the combination of a very high number of civilian-targeting events and high lethality. Based on these two indicators, it shows the highest Relative Attention Index, highlighting a particularly severe concentration of violence and reported human loss affecting civilians.

The event map also reveals two main spatial concentrations of civilian-targeting events:

- **air/drone strikes** are concentrated mainly in **Gaza**
- **attacks** are more widespread across the **West Bank**

# Which countries show the greatest need for attention from a civilian protection perspective?

GLOBAL VIEW



The **Democratic Republic of Congo** also shows a particularly severe combination of **high civilian-targeting frequency and high lethality**. With **7,908 events** and **17,741 reported fatalities**, it records the **second-highest relative attention index** in the analysis.

The event map shows that civilian-targeting events are **not evenly distributed across the country**, but are concentrated mainly in the **eastern part of the DRC**, especially along the border areas.

The spatial pattern also reveals that:

- **attacks** are widely distributed across the main affected eastern areas
- **abductions/forced disappearances** are also strongly represented and often overlap with areas where attacks are concentrated

## Conclusions

The analysis shows that **civilian-targeting violence is highly uneven across the world**, with a limited number of regions and countries accounting for a large share of recorded events. However, **frequency alone does not explain the human cost**: regions with fewer events can still experience very high reported fatalities, showing that the severity of violence also depends on its form.

Different types of civilian-targeting produce very different levels of reported human loss, and these patterns vary across regions. The most severe situations therefore emerge where **high event frequency and high lethality overlap**. In this analysis, **Palestine and the Democratic Republic of Congo** stand out on both dimensions and record the highest Relative Attention Index values.

Overall, the project highlights that **understanding civilian harm requires looking beyond how often violence occurs**, and considering together **where it happens, how much human loss it causes, and which forms of violence are associated with the greatest reported fatalities**.

## LICENCE

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